■ Future Scope & Conclusion

Improvements, expansions, and project summary

Project Conclusion

The Automatic Exhaust Fan project successfully demonstrates an innovative solution for smart air quality management. By integrating the MQ-2 gas sensor with the ESP8266 NodeMCU and a relay module, the system automatically controls exhaust fan operation based on real-time smoke detection — without any manual intervention.

The project achieves its core objectives: energy efficiency (fan runs only when needed), health protection (removes smoke and harmful gases), fire hazard prevention, and reduced noise pollution. The system was tested and validated through the Arduino Serial Monitor and ThingSpeak IoT dashboard.

Key Achievements

- ■ Automatic smoke detection and fan activation without manual input
- ■ Energy savings — fan operates only when smoke is present
- ■ Multi-level LED indication (Green / Orange / Red) for air quality status
- ■ Buzzer alarm for immediate hazard notification
- ■ Real-time data logging on ThingSpeak IoT platform
- ■ Low-cost implementation using widely available components
- ■ Simple and reliable code with adjustable threshold (thresh = 500)

Future Scope & Improvements

1. Multi-Sensor Integration

- Add DHT11/DHT22 for temperature & humidity monitoring
- Add MQ-7 for Carbon Monoxide (CO) detection
- Add MQ-135 for CO_2, NH_3, and VOC detection
- Add PM2.5 particulate matter sensor for fine dust monitoring

2. Fan Speed Control (PWM)

- Replace ON/OFF relay with PWM motor driver (L298N or L293D)
- Vary fan speed based on smoke intensity level
- Low speed for minor smoke, full speed for heavy smoke
- Saves energy by not running at full power unnecessarily

3. Mobile App Notifications

- Use ESP8266 Wi-Fi to send push notifications via Blynk or Firebase
- SMS alerts using Twilio API or SIM800L GSM module

• Real-time dashboard showing smoke levels on a mobile app

4. Noise Reduction

• Integrate noise sensor to monitor fan sound levels
• Use quieter brushless DC fans (BLDC)
• Add soundproofing material around fan housing

5. Building Management System (BMS) Integration

• Connect to centralized BMS for large buildings, hotels, malls
• Integrate with existing HVAC systems
• MQTT protocol for IoT communication with central server

6. Solar Power Integration

• Add a solar panel and battery backup for power outages
• Makes the system fully autonomous and off-grid capable

7. Voice Assistant Integration

• Connect with Amazon Alexa or Google Home for voice control
• Ask 'Is the kitchen safe?' and get real-time air quality response

References

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